The positive difference space need not be closed: a Hilbert lattice-cone counterexample

New counterexample for arXiv:1501.06696

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Status. Full negative answer, in a strengthened form. There is a real Hilbert ordered gradient space for which $V_0 = V_+ - V_+$ is a proper dense subspace of V, hence is not closed. Moreover, both V_+ and V_0 are lattices, so the counterexample also fits the lattice setting immediately surrounding the source question.

Source question

Arnlind, Björn and Björn define

$$V_0 = \{u \in V : u \leq v \text{ for some } v \in V_+\} = V_+ - V_+.$$

After proving that this is a linear subspace, they state that they do not know whether it is always closed [1, p. 19].

$u_1, u_2 \in V_+$, their V_+-least upper bound $\max(u_1, u_2)$ *is even least among all upper bounds in V.*

Similarly, the V_+-greatest lower bound $\min(u_1, u_2)$ *is greatest also among all lower bounds of u_1 and u_2 in V.*

In particular, it follows that one can replace "V_+-least" and "V_+-greatest" by "V-least" and "V-greatest" (with the obvious interpretation) in Theorem 6.15 (while leaving the other V_+'s) to produce four more equivalent statements.

Theorem 6.17. *V is a lattice if and only if $V = V_0$ and V_+ is a lattice.*

Note that $V = V_0$ if and only if every element in V has an upper bound in V_+, or equivalently, if every pair of elements in V has a common upper bound in V.

The condition $V = V_0$ cannot be dropped, as seen by the following example.

Example 6.18. Let $V = \widetilde{V} = \mathbf{C}$ and say that $u \leq v$ if $\operatorname{Re} u \leq \operatorname{Re} v$ and $\operatorname{Im} u = \operatorname{Im} v$. Then $V_+ = \{\lambda \in \mathbf{R} : \lambda \geq 0\}$ and $V_0 = \mathbf{R} \neq V$. Here V_+ and V_0 are lattices, but V is not.

We do not know if V_0 is always closed, but we next show that V_0 is indeed a linear subspace.

Source evidence: arXiv:1501.06696, p. 19.

Proof intuition

In the n-th copy of $\mathbb{R}^2$, take the positive wedge generated by two vectors that become almost parallel:

$$p_n = (1, 0), \qquad q_n = (1, 1/n).$$

The blockwise direct sum of these wedges is a closed positive cone in a Hilbert space. Yet writing a vector as a difference of positive vectors requires its coordinates in the increasingly ill-conditioned bases (p_n, q_n) to remain square summable. This produces a dense graph-domain inside the Hilbert space rather than a closed subspace. The particularly transparent witness has blocks $q_n - p_n = (0, 1/n)$: the vector is square summable, but its coefficient sequences are the nonsummable constants (-1) and 1.

The wedges are acute, so adding a positive vector cannot decrease the Hilbert norm. Their basis coordinates also make maxima and minima coordinatewise, which preserves the lattice property.

Theorem 1. *There exists a real ordered gradient space $\mathcal{U} = (V, \widetilde{V}, W, \widetilde{W}, R)$ such that its positive cone V_+ is a lattice and*

$$V_0 = V_+ - V_+$$

is a proper dense, nonclosed vector subspace of V. The space V can be chosen to be a Hilbert space, and V_0 itself is a vector lattice.

Proof. Let

$$H = \ell^2(\mathbb{N}; \mathbb{R}^2)$$

with its usual real Hilbert norm. For $n \geq 1$, set

$$p_n = (1, 0), \qquad q_n = (1, 1/n), \qquad C_n = \{a p_n + b q_n : a, b \geq 0\} \subset \mathbb{R}^2,$$

and define

$$K = \{z = (z_n) \in H : z_n \in C_n \text{ for every } n\}.$$

Each C_n is a closed, convex, pointed cone. Since the coordinate projections $H \to \mathbb{R}^2$ are continuous, K, their inverse-image intersection, is also a closed, convex, pointed cone. It defines a partial order on H by $u \leq v$ exactly when $v - u \in K$.

This order is compatible with the Hilbert norm. Indeed, every two vectors of C_n have nonnegative inner product: all four inner products among the generators p_n, q_n are positive. Hence $\langle u, w \rangle_H \geq 0$ for $u, w \in K$. If $0 \leq u \leq v$, write $v = u + w$ with $w \in K$; then

$$\|v\|_H^2 = \|u\|_H^2 + 2\langle u, w \rangle_H + \|w\|_H^2 \geq \|u\|_H^2.$$

We now compute $K - K$. Every block has unique (real) coordinates

$$z_n = r_n p_n + s_n q_n = (r_n + s_n, s_n/n), \qquad s_n = n z_{n,2}, \quad r_n = z_{n,1} - n z_{n,2}. \tag{1}$$

If $k \in K$ has nonnegative coefficients a_n, b_n, then $k_{n,1} = a_n + b_n$. Thus $k \in H$ implies $(a_n), (b_n) \in \ell^2$. It follows that if $z = k - \ell \in K - K$, then both coefficient sequences $(r_n), (s_n)$ in (1) belong to ℓ^2. Conversely, if $r, s \in \ell^2$, splitting each into positive and negative parts writes

$$z = (r^+ p + s^+ q) - (r^- p + s^- q)$$

as the difference of two elements of K. Consequently,

$$K - K = \{z \in H : (r_n), (s_n) \in \ell^2\} = \{z \in H : (n z_{n,2}) \in \ell^2\}. \tag{2}$$

The last equality uses $(z_{n,1}) \in \ell^2$.

The right-hand side of (2) contains every finitely supported vector, so it is dense in H. It is not all of H. Namely, let

$$z_n^* = (0, 1/n) = q_n - p_n.$$

Then $z^* \in H$, since $\sum_n n^{-2} < \infty$, but its unique coefficient sequences are $r_n = -1$ and $s_n = 1$, which are not in ℓ^2. Thus $z^* \notin K - K$. Its finite truncations do belong to $K - K$ and converge to z^*, explicitly witnessing nonclosedness.

For completeness, the construction preserves lattice structure. In the basis (p_n, q_n), the order in every block is coordinatewise. A positive element of K has two nonnegative ℓ^2-coefficient sequences, so coordinatewise maxima and minima of two such elements again lie in K. Thus K is a lattice. Formula (2) likewise shows that maxima and minima of the coefficient pairs of two elements of $K-K$ remain in ℓ^2; hence $K-K$ is a vector lattice.

Finally, turn this ordered Hilbert space into an ordered gradient space in the source's sense. Take

$$V = \widetilde{V} = W = \widetilde{W} = H, \qquad R = \{(u,u) : u \in H\}.$$

The identity relation is a closed linear gradient relation; the Hilbert spaces are reflexive and strictly convex. The preorder is compatible with norm limits because K is closed, and the required norm monotonicity was proved above. Therefore this is an ordered gradient space. Its positive cone is $V_+ = K$, and its source-defined subspace is $V_0 = V_+ - V_+ = K - K$, which is dense and nonclosed by the preceding calculation. □

Verification and literature status

The bundled exact-arithmetic script checks the block-coordinate inverses, acuteness on an exhaustive finite grid, the identity $q_n - p_n = (0, 1/n)$, its coefficient pair $(-1, 1)$, and convergence of finite truncations. These are independent sanity checks; the proof above is complete without computation.

Exact-phrase, title, author, and ordered-gradient-space searches through 13 August 2026 found no later arXiv source answering this question. This was a bounded search rather than a systematic bibliographic review.

Reference

[1] J. Arnlind, A. Björn and J. Björn, *An axiomatic approach to gradients with applications to Dirichlet and obstacle problems beyond function spaces*, Commun. Contemp. Math. **18** (2016), 1550060; arXiv:1501.06696.